Graph Contrastive Learning With High-Order Relations for Hyperspectral Image Clustering

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***Abstract*—Hyperspectral image (HSI) clustering has shown remarkable potential in Earth observation, as it can utilize the rich spectral information for classification without depending on labels. Deep clustering methods based on contrastive learning have drawn significant attention from researchers due to their high performance. However, existing contrastive learning-based methods are unable to capture high-order positive samples and frequently misclassify negative samples. To address this issue, we propose a novel graph contrastive learning method with high-order relations (GCLHR) for HSI clustering. Specifically, we construct an affinity matrix and employ multiple random walks to capture the distant positive samples of each sample while avoiding capturing false negative samples. Meanwhile, the similarity measurement also filters out the false negative samples to a certain extent. Before formal training, we use a 3-D autoencoder-based pretraining to capture deep spectral and spatial features to provide the input of the graph convolutional network. Moreover, GCLHR utilizes clustering semantic information to eliminate noisy edges for view augmentation and designs a triplet self-supervised learning module to enhance the consistency between views. Experiments carried out on multiple HSI datasets demonstrate that our proposed GCLHR method attains excellent and reliable clustering performance.**

***Index Terms*—Contrastive learning, deep clustering, graph convolutional network, hyperspectral images (HSIs).**

NOMENCLATURE

k	Number of neighbors.
M	Number of superpixels.
K	Number of clusters.
τ	Temperature coefficient.
N	Number of samples.
$D, d, \hat{d}, \tilde{d}$	Dimension of data.
$\mathbf{I} \in \mathbb{R}^{M\times M}$	Identity matrix.
$\mathbf{X} \in \mathbb{R}^{W\times H\times D}$	Input hyperspectral image data.
$\mathbf{X}^{sp} \in \mathbb{R}^{M\times D}$	Set of superpixels.
$\mathbf{D} \in \mathbb{R}^{M\times M}$	Degree matrix of graph.
$\mathbf{P} \in \mathbb{R}^{N\times w\times w\times d}$	Pixel cubes.
$\hat{\mathbf{P}} \in \mathbb{R}^{N\times w\times w\times d}$	Reconstructed pixel cubes.
$\mathbf{X}^{p} \in \mathbb{R}^{N\times \tilde{d}}$	Deep features extracted by VAE.
$\mathbf{Z} \in \mathbb{R}^{M\times \tilde{d}}$	Initial features of superpixels.
$\mathbf{A}^{v_1}, \mathbf{A}^{v_2} \in \mathbb{R}^{M\times M}$	Constructed adjacency matrices.
$\tilde{\mathbf{A}}^{v_1}, \tilde{\mathbf{A}}^{v_2} \in \mathbb{R}^{M\times M}$	Normalized adjacency matrices.
$\mathbf{H}^{v_1}, \mathbf{H}^{v_2} \in \mathbb{R}^{M\times \hat{d}}$	Deep features extracted by GCN.
$\mathbf{U}^{t} \in \mathbb{R}^{M\times M}$	Affinity matrix after t propagations.
$\mathbf{M} \in \mathbb{R}^{M\times M}$	Transition matrix of the random walk.
$\|\cdot\|_2$	L2 norm.
$\|\cdot\|_F$	Frobenius norm.
$\mathcal{H}(\cdot\|\|\cdot)$	Cross-entropy function.
$\mathrm{KL}(\cdot\|\|\cdot)$	Kullback–Leibler divergence.
$\mathbf{\Theta}_{\mathrm{VAE}}$	Encoding parameter of VAE.
$\mathbf{\Omega}_{\mathrm{VAE}}$	Decoding parameter of VAE.
$\mathbf{\Theta}_{\mathrm{GCN}}$	Encoding parameter of GCN.

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I. INTRODUCTION

HYPERSPECTRAL images (HSI) are rich in spectral information, which can effectively support the task of Earth observation [1], [2], [3] and are widely used in fields such as object tracking [4], [5], mining area categorization [6], and image segmentation [7]. However, the HSI processing task requires a large number of labels for model training, which demands significant human resources [8]. To alleviate this situation, the unsupervised HSI clustering task has emerged.

HSI clustering aims to obtain discriminative features in order to classify each pixel into different clusters without relying on labels [9], [10], [11], [12]. Traditional methods usually achieve clustering by analyzing inherent data characteristics or transforming clustering tasks into optimization [13], [14]. Common methods include centroid-based [15], [16], graph-based [17], [18], [19], [20], and subspace-based methods [21], [22], [23]. The advantage of traditional methods is that they consume fewer computing resources and can efficiently solve simple clustering tasks. Nevertheless, since traditional methods are unable to extract more discriminative deep features, they

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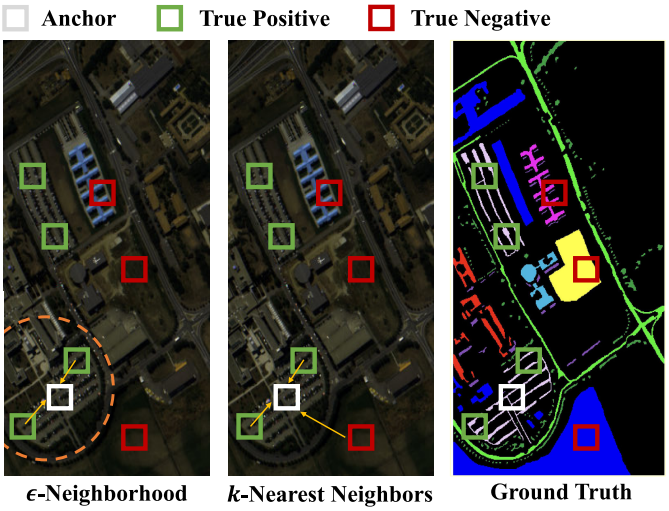


Fig. 1. In ϵ- and k-neighbor-based methods for constructing positive and negative sample pairs, all samples within a fixed range are usually regarded as positive, while those beyond the range are considered negative. Such a rigid criterion, however, can result in false negative and false positive samples.

do not perform well in complex tasks with high heterogeneity [24], [25].

Deep learning methods are capable of uncovering more discriminative features within HSI [26], [27], [28], and existing deep clustering methods can generally be divided into two categories: autoencoder-based [29], [30], [31], [32] and contrastive learning-based methods [33], [34], [35], [36], [37]. The autoencoder-based approaches aim to reconstruct features to be as similar as possible to the input data with constraints to avoid overfitting, thereby enhancing the model's feature extraction capabilities. To capture long-range spatial information, researchers have developed a graph autoencoder that utilizes neighboring information by integrating structural knowledge [38]. The contrastive learning-based methods focus on distinguishing negative samples while pushing positive samples closer [39]. Therefore, the design of positive and negative sample pairs is of great importance in contrastive learning tasks. The existing contrastive learning methods typically adopt two ways to construct positive and negative sample pairs for HSI clustering tasks. One way is to construct a new view by data augmentation and choose the corresponding sample as the positive sample, with the remaining samples regarded as negative samples [36], [40], [41]. The other way is to choose the nearest k neighbors of each sample as positive samples while considering the rest of the samples as negative samples [33], [35].

Although these simple methods for constructing positive and negative sample pairs show promising performance, they may raise false positive and false negative issues. As shown in Fig. 1, the ϵ- and k-neighbor-based methods for constructing positive and negative sample pairs generally consider only first-order neighbors as positive samples, neglecting information from distant neighborhoods. This omission not only incorporates heterogeneous out-of-cluster samples, such as background information, but also misclassifies a significant number of same-cluster samples as negative, resulting in class conflict issues. Moreover, existing graph contrastive learning algorithms usually employ edge perturbations or add noise information to construct data augmentation branches. However, this noise information cannot be eliminated during training and will interfere with model training. These challenges are particularly severe in HSI and other remote sensing data due to their unique characteristics. HSI data exhibit high spectral dimensionality, where spectrally similar signatures may belong to distinct land-cover classes, and strong spatial heterogeneity, where pixels within the same object may vary substantially because of illumination, shadow, or atmospheric effects. Hence, it is of great significance to develop an algorithm that can effectively alleviate the false positive and false negative problems in graph contrastive learning with efficient data augmentation branches.

Based on the above discussion, we propose a novel graph contrastive learning method with high-order relations (GCLHR) for HSI clustering, which utilizes random walks to capture high-order similarity relationships between samples. In detail, GCLHR encompasses four important modules, namely, pretraining based on 3-D autoencoder, graph data augmentation based on semantic information, graph contrastive learning with high-order relations, and triplet self-supervised learning. First, we employ 3-D autoencoders to extract deep spatial and spectral features, which lay a foundation for subsequent superpixel segmentation and graph convolutional networks (GCNs). Subsequently, we utilize superpixel segmentation methods to divide the data into irregular superpixels and design a distance-based superpixel feature calculation method instead of simply averaging pixel features. Meanwhile, to avoid the noise problem introduced by random augmentation, we adopt a semantic-based structural augmentation method to remove incorrectly connected samples and achieve data augmentation. Finally, contrastive learning loss and self-supervised loss are used to assist GCN in extracting clustering-friendly deep node representations. Contrastive learning with high-order relations represents the similarity between nodes by calculating affinity graphs, and multiple random walks are used to calculate relationships of high-order similarity for each node. Such a capture of long-distance relationships can alleviate the problems of false positive samples and false negative samples to the greatest extent possible. In addition, a well-designed triplet self-supervised clustering objective is used to obtain the cluster-oriented output embedding. Our main contributions are summarized as follows.

1) We propose a novel contrastive learning framework for HSI clustering that uses a random walk approach to obtain the high-order relations so that the correct and distant positive samples can be discovered.
2) A graph data augmentation method based on semantic information and a triplet self-supervised learning approach are designed to help the network learn clustering-friendly features.
3) Extensive experiments across four widely used HSI datasets have demonstrated the effectiveness and superiority of the proposed method.

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II. Related Work

A. GCN-Based HSI Clustering

GCN has found extensive application in HSI clustering tasks, as it is capable of aggregating neighboring information by leveraging structural knowledge [42], [43], [44]. Existing approaches can be principally categorized into traditional methods and deep learning methods. Traditional methods predominantly rely on extracting data features through a simple graph convolution for optimization, under the premise that these methods consider the activation function of graph convolution to be of secondary importance [45], [46]. Cai et al. [47] were the first to propose a GCN-based method for extracting the spatial information of HSI and integrating it with a subspace clustering method for self-expression, thereby capturing the interdata relationships. Building upon this, Liu and Wang [48] incorporated the optimal transport method to capture the intrinsic geometric structure information among features of HSI and achieved certain outcomes. Liu et al. [9] integrated multiview information of HSI and adaptive graph structures to mitigate the impact of unreasonable neighbor relationships on the model.

Although these methods can attain certain results, they fall short in the in-depth exploration of data features [49], [50], [51], which has led to the growing emphasis on deep GCN-based clustering methods. Deep GCN can typically be formulated as

$$\mathbf{H}^{(l+1)} = \psi\left(\mathbf{H}^{(l)}\tilde{\mathbf{A}}\mathbf{W}^{(l)}\right) \tag{1}$$

where $\mathbf{H}^{(0)}$ is the initial data, $\mathbf{H}^{(l)}$ is the learned feature, and l is the layer of GCN. $\mathbf{W}^{(l)}$ is the linear transformation matrix and $\psi(\cdot)$ is an activation function. Zhang et al. [29] devised a dual autoencoder for extracting the spectral and spatial features of HSI. Han et al. [38] combined 1-D graph autoencoders and 2-D autoencoders to extract pixel features and pixel patch features of the HSI data, respectively, and then fused them to obtain diverse information. Peng et al. [52] designed a graph autoencoder guided by structural knowledge, with the focus on reconstructing the adjacency structure of the HSI data, thereby acquiring effective high-order structural information. Ding et al. [53] designed a GCN structure with low-pass characteristics and combined it with self-supervisory information to conduct HSI clustering. Although these methods can effectively extract deep features, they continue to employ simple neighbor-based approaches for graph construction, which inevitably introduce a considerable number of heterogeneous edges and result in suboptimal performance. To overcome this limitation, we propose a graph enhancement strategy guided by semantic performance, aiming to generate a more accurate and optimal graph structure.

B. Contrastive Learning-Based HSI Clustering

Contrastive learning endeavors to separate negative samples and draw in positive samples, thereby facilitating the model in extracting more discriminative features [40], [54]. Consequently, the crux of contrastive learning resides in the design of positive and negative sample pairs. The classical contrastive learning loss [55] is formulated as follows:

$$\mathcal{L}\left(\mathbf{h}_i^1, \mathbf{h}_i^2\right) = -\log \frac{\mathrm{e}\left(\theta\left(\mathbf{h}_i^1, \mathbf{h}_i^2\right)/\tau\right)}{\mathrm{e}\left(\theta\left(\mathbf{h}_i^1, \mathbf{h}_i^2\right)/\tau\right) + \sum_{j\neq i}\mathrm{e}\left(\theta\left(\mathbf{h}_i^1, \mathbf{h}_j^2\right)/\tau\right)} \tag{2}$$

where the sample pair $\mathbf{h}_i^1$ and $\mathbf{h}_i^2$ represents the same sample of the augmented view, i.e., the positive sample. The sample pair $\mathbf{h}_i^1$ and $\mathbf{h}_j^2$ represents different samples of the enhanced view, i.e., the positive sample. θ is the cosine similarity and τ represents the temperature coefficient.

Owing to its facile integration and potent performance, it has also gained prevalence in the domain of HSI clustering. Guan et al. [33] designed pixel-level and superpixel-level contrastive learning (PSCPC) to capture a variety of information. Cai et al. [35] put forward the nearest neighbor contrastive subspace clustering method (NCSC), which conducts contrastive learning by regarding the nearest k samples of each sample as positive samples and the remaining samples as negative samples. Qi et al. [56] contrived a semantic-aware data augmentation method and a contrastive learning approach based on cluster centers to mitigate the issue of category conflict. Luo et al. [34] employed distinct encoders to extract diverse features for contrastive learning and incorporated the joint self-supervisory loss to optimize the model. Guan et al. [36] proposed a multiview-based contrastive learning method that extracted texture information from HSI to construct novel views for contrastive learning and also incorporates new view information. Wang et al. [41] designed multilevel contrastive learning and utilized the encoding information of different layers to conduct multiple contrastive learning. However, these methods for constructing positive and negative sample pairs generally consider only first-order neighbors as positive samples, neglecting information from distant neighborhoods. Thus, we propose a graph contrastive learning method based on high-order relations to solve this problem.

III. Methods

In this section, we introduce the proposed GCLHR framework for HSI clustering. As depicted in Fig. 2, GCLHR is composed of four core modules: (a) a 3-D autoencoder–based pretraining module, (b) a semantically guided graph augmentation and encoding module, (c) a high-order relation-driven contrastive learning module, and (d) a triple self-supervised learning module. The following subsections provide detailed descriptions of each component.

A. Notations

Given an HSI cube $\mathbf{X} \in \mathbb{R}^{W\times H\times D}$ containing $N = W \times H$ pixels and D spectral bands, where W and H are the length and width of the data, respectively, to reduce the amount of calculation, we divide it into superpixels to obtain M superpixels. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the corresponding arbitrary undirected graph, where $\mathcal{V} = \{v_1, \ldots, v_n\}$ and $\mathcal{E}$ are, respectively, the node set and edge set. We use the superpixel-level node feature matrix $\mathbf{X}^{sp} \in \mathbb{R}^{M\times d}$ and adjacency matrix $\mathbf{A} \in \mathbb{R}^{M\times M}$ to

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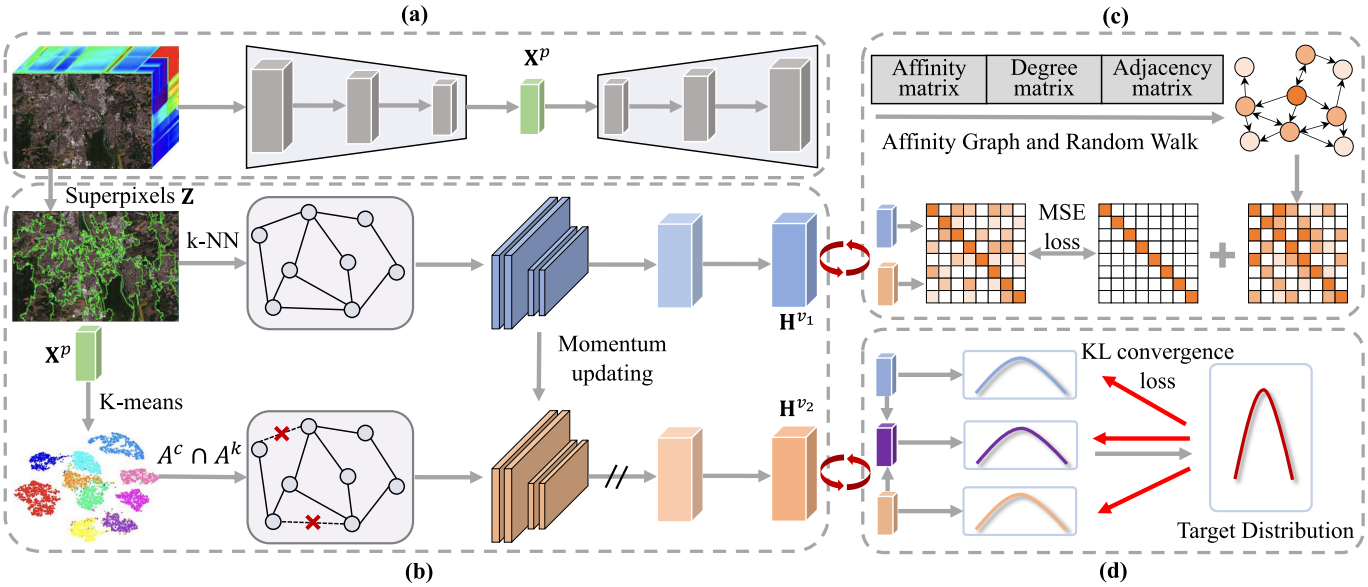


Fig. 2. Illustration of the proposed framework GCLHR. Within this model, four modules are included. (a) 3-D Autoencoder-based pretraining, (b) semantic information-based graph augmentation, (c) contrastive learning with high-order relations, and (d) triplet self-supervised learning.

characterize this graph, where $\mathbf{X}_i^{sp}$ is the feature vector of node v_i, and $a_{ij} = 1$ if $(v_i, v_j) \in \mathcal{E}$. The normalized adjacency matrix is denoted as $\tilde{\mathbf{A}} = \hat{\mathbf{D}}^{-1/2}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-1/2}$, where $\hat{\mathbf{A}}$ equals $\mathbf{A} + \mathbf{I}$, where $\mathbf{I}$ is the identity matrix and $\hat{\mathbf{D}}$ is the diagonal degree matrix. All frequently used notations are listed in Nomenclature.

B. Pretraining Based on 3-D Autoencoder

Existing HSI clustering algorithms usually divide the original data into superpixels or patches to directly extract deep features, but these methods ignore the 3-D structural information of HSI and thus fail to jointly utilize spectral information and spatial information. Inspired by existing literature [56], we use a 3-D autoencoder to preprocess HSI, where the encoder contains two layers of 3-D-convolutional neural network (CNN) and one layer of 2-D-CNN, and the decoder has the opposite structure. Next, we use the principal component analysis method to reduce the dimension of the HSI into d dimensions, because the HSI contains a lot of redundant spectral information. Then, we use the sliding window method to extract the surrounding $w \times w$ patches for each pixel of the HSI to construct pixel cubes $\mathbf{P}_i \in \mathbb{R}^{w \times w \times d} (i = 1, 2, \ldots, N)$, where $\mathbf{P}_i$ is the ith pixel cube. We feed the pixel cubes $\mathbf{P}$ into the encoder for feature extraction and use the fully connected layer to reduce the dimensionality. After we obtain the mean vector $\boldsymbol{\mu}$ and the covariance $\boldsymbol{\sigma}$, we use the following equation to obtain the latent vector:

$$\mathbf{q} = \boldsymbol{\mu} + \Upsilon \times \boldsymbol{\sigma} \tag{3}$$

where Υ is sampled from the standard normal distribution. According to the mean vector $\boldsymbol{\mu}$ and standard deviation vector $\boldsymbol{\sigma}$, we can construct distribution loss

$$\mathcal{L}_{\boldsymbol{\mu},\boldsymbol{\sigma}^2} = \frac{1}{2}\sum_{i=1}^{\hat{d}} \left(\boldsymbol{\mu}_{(i)}^2 + \boldsymbol{\sigma}_{(i)}^2 - \log \boldsymbol{\sigma}_{(i)}^2 - 1\right). \tag{4}$$

The deep features $\mathbf{X}^p$ are then decoded into the reconstructed features $\overline{\mathbf{P}}_i$ using a decoder with the opposite structure, so the reconstruction loss can be expressed as

$$\mathcal{L}_{\text{rec}} = \frac{1}{2}\sum_{i=1}^{N} \left(\left\|\mathbf{P}_i - \overline{\mathbf{P}}_i\right\|_F^2\right) \tag{5}$$

where $\|\cdot\|_F$ denotes the Frobenius norm function. The total loss of pretraining is

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\boldsymbol{\mu},\boldsymbol{\sigma}^2} + \mathcal{L}_{\text{rec}}. \tag{6}$$

C. Homogeneous Region Generation

Considering the large amount of remote sensing data, in order to reduce the computational complexity, we use the entropy rate superpixel segmentation (ERS) [57] method to segment the HSI into superpixels $\mathcal{S} = \bigcup_{i=1}^{M} \mathbf{x}_i^{sp}$, where $\mathbf{x}_i^{sp} = 1/N_i \sum_{j=1}^{N_i} (\mathbf{x}^p)_j^i$ represents the ith superpixel, $(\mathbf{x}^p)_j^i$ denotes the jth element in $\mathbf{x}_i^{sp}$, M is the number of superpixels, N_i is the number of pixel in $\mathbf{x}_i^{sp}$, $\mathbf{x}_i^{sp} \cap \mathbf{x}_j^{sp} = \emptyset \ \forall i \neq j$, and $\sum_{i=1}^{M} N_i = N$. The number of superpixels after segmentation is much smaller than the number of pixels, so we can reduce the amount of computation.

However, previous methods usually average the features of pixels within a superpixel to obtain the superpixel features. The calculation is usually as follows:

$$\boldsymbol{\delta}_i = \frac{1}{N_i}\sum_{j=1}^{N_i} (\mathbf{x}^p)_j^i. \tag{7}$$

However, the simple and crude average ignores the importance of each pixel. To this end, we design the correlation matrix to measure the importance of each pixel in the superpixel. First, we need to calculate the covariance matrix $\boldsymbol{\Gamma}$

$$\boldsymbol{\Gamma}_i = \frac{1}{N_i - 1}\sum_{j=1}^{N_i} \left((\mathbf{x}^p)_j^i - \boldsymbol{\delta}_i\right)\left((\mathbf{x}^p)_j^i - \boldsymbol{\delta}_i\right)^{\mathrm{T}}. \tag{8}$$

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We design a distance-based mapping mechanism based on the $\boldsymbol{\delta}$ and $\boldsymbol{\Gamma}$. The correlation matrix $\mathbf{O}$ is designed to transfer the spatial correlation between features and superpixels

$$\mathbf{O}_{i,j} = \begin{cases} \sqrt{\left((\mathbf{x}^p)_i - \boldsymbol{\delta}_j\right)^{\mathrm{T}} \boldsymbol{\Gamma}_j^{-1} \left((\mathbf{x}^p)_i - \boldsymbol{\delta}_j\right)}, & \text{if } \mathbf{x}^p \in \mathbf{x}^{sp} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $\mathbf{O}$ describes the distance between pretraining feature $\mathbf{x}^p \in \mathbf{x}^{sp}$ and the centroid of $\mathbf{x}^{sp}$ in a Mahalanobis space. We intentionally preserve this positive correlation between distance and weight to strengthen the response to pixels located at superpixel boundaries, enabling the aggregated superpixel features to better capture boundary variations

$$\mathbf{Z} = \tilde{\mathbf{O}}^{\mathrm{T}} \mathbf{X}^p, \quad \tilde{\mathbf{O}}_{i,j} = \mathbf{O}_{i,j} / \sum_m \mathbf{O}_{m,j} \tag{10}$$

where $\mathbf{Z}$ is the superpixel feature we calculated and $\tilde{\mathbf{O}}$ is the normalized $\mathbf{O}$.

D. Semantic Information-Based Graph Augmentation

Since these superpixels are features at the vector level, we employ graph structures to represent them. Given that the propagation of graph structures hinges on the adjacency matrix, we utilize a common distance algorithm to capture the k nearest neighbors for each sample in order to construct the adjacency matrix

$$\mathbf{A}_{ij}^{v_1} = \begin{cases} 1, & \text{if } \mathbf{z}_j \in \Gamma(\mathbf{z}_i) \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where $\Gamma(\cdot)$ represents the top k nearest samples set of sample.

The construction process of the adjacency matrix $\mathbf{A}^{v_1}$ roughly selects an equal number of neighbors for each sample. This approach is not suitable for real-world applications and overlooks the semantic information among samples. Moreover, the previous algorithm randomly interferes with the edges of the adjacency matrix, which easily introduces noise information. For this reason, we devise a semantically enhanced graph data augmentation method. Specifically, we apply the K-means clustering algorithm to the feature $\mathbf{Z}$ to obtain a cluster structure, where the number of clusters is equal to the number of true labels. Subsequently, samples within the same cluster are regarded as connected neighbors; otherwise, they are considered unconnected. In this way, the global semantic association matrix $\mathbf{A}^c$ is obtained. Considering the instability of K-means, we utilize the intersection of multiple clusterings to acquire reliable global information and derive an enhanced adjacency matrix $\mathbf{A}^{v_2}$

$$\begin{aligned} \mathbf{A}^c &= \mathbf{A}^{c1} \cap \mathbf{A}^{c2} \cap \cdots \cap \mathbf{A}^{cn} \\ \mathbf{A}^{v_2} &= \mathbf{A}^c \cap \mathbf{A}^{v_1}. \end{aligned} \tag{12}$$

Based on the two adjacency matrices obtained, we construct two graph views $\mathcal{G}^{v_1} = (\mathcal{V}, \mathcal{E}^{v_1})$ with $\{\tilde{\mathbf{A}}^{v_1}, \mathbf{Z}\}$ and $\mathcal{G}^{v_2} = (\mathcal{V}, \mathcal{E}^{v_2})$ with $\{\tilde{\mathbf{A}}^{v_2}, \mathbf{Z}\}$. Then, to aggregate the long-range spatial information between nodes, we use a multilayer graph convolutional network to extract deep features

$$\mathbf{H}^{v_1} = f\left(\widetilde{\mathcal{G}}^{v_1} \mid \Theta_{\mathrm{GCN}}^{v_1}\right) = f\left(\widetilde{\mathbf{A}}^{v_1}, \mathbf{Z} \mid \Theta_{\mathrm{GCN}}^{v_1}\right) \tag{13}$$

$$\mathbf{H}^{v_2} = f\left(\widetilde{\mathcal{G}}^{v_2} \mid \Theta_{\mathrm{GCN}}^{v_2}\right) = f\left(\widetilde{\mathbf{A}}^{v_2}, \mathbf{Z} \mid \Theta_{\mathrm{GCN}}^{v_2}\right) \tag{14}$$

where $\Theta_{\mathrm{GCN}}^{v_1}$ and $\Theta_{\mathrm{GCN}}^{v_2}$ denote the learnable parameters of two graph encoders.

E. Contrastive Learning With High-Order Relation

The existing neighbor-based positive and negative sample selection method generally considers samples with adjacent relationships as positive samples. This approach is prone to false positive issues, namely, samples outside the cluster are misidentified as positive samples. To avoid this problem, we design a high-order contrastive learning method based on random walks. In this method, we strive to consider samples within the cluster as positive samples and assign them higher weights, while regarding samples outside the cluster as negative samples. Specifically, we conduct random walk operations on the graph. Based on the weights between nodes, we calculate the probabilities of positive and negative samples. Multiple random walk operations can capture positive samples that are at a long distance from the nodes, thus ensuring a larger number of positive samples. The transition matrix of the random walk can be expressed as

$$\mathbf{M} = \mathbf{U}\mathbf{D}^{-1} \tag{15}$$

where $\mathbf{U}$ denotes the affinity matrix, $\mathbf{U}_{ij}$ is the weight of between node i and j, $\mathbf{D} = \operatorname{diag}(d_1, d_2, \ldots, d_M)$, and $d_i = \sum_{j=1} \mathbf{U}_{ij}$.

The affinity matrix is fully connected and can reflect the similarity relationship between each node. It is defined as follows:

$$\mathbf{U}_{ij} = \exp\left(-\left\|\mathbf{z}_{i,l} - \mathbf{z}_{j,l}\right\|^2 / \sigma\right) \tag{16}$$

where $\mathbf{z}_{i,l}$ represents the lth channel of the feature and σ is a scaling parameter that is fixed to 0.1 in all experiments. In order to minimize the false positive problem, we incorporate clustering semantic information into it, that is, we select the corresponding value in $\mathbf{U}_{ij}$ based on the nonzero elements in $\mathbf{A}^c$.

As previously mentioned, to acquire discriminative relationships over longer distances, we conduct t random walks to obtain the transition matrix $\mathbf{M}^t$. $\mathbf{M}^t$ contains the connection probability between samples and can be regarded as soft labels between samples, so we can implement an enhanced contrastive learning algorithm based on $\mathbf{M}^t$. In addition, because the sample itself must be a positive sample, we add a higher weight to each sample itself

$$\hat{\mathbf{M}}^t = \alpha \mathbf{I}_n + (1 - \alpha)\mathbf{M}^t \tag{17}$$

where α is a tradeoff parameter that is fixed to 0.5. In this manner, the transition matrix that combines its own and long-distance discriminative information can serve as an excellent target for contrastive learning. Thus, our improved contrastive learning with high-order relations can be expressed as

$$\mathcal{L}_{\mathrm{chr}} = \mathcal{H}\left(\hat{\mathbf{M}}^t, \rho\left(\mathbf{h}^{v_1}, \mathbf{h}^{v_2}\right)\right) \tag{18}$$

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where $\mathcal{H}(\cdot\|\cdot)$ denotes the cross-entropy function and $\rho(\cdot\|\cdot)$ is the similarity with the row-wise normalization manner

$$\rho\left(\mathbf{h}_i^{v_1}, \mathbf{h}_j^{v_2}\right) = \frac{\mathrm{e}\left(\theta\left(\mathbf{h}_i^{v_1}, \mathbf{h}_j^{v_2}\right)/\tau\right)}{\sum_{t=1}^{M} \mathrm{e}\left(\theta\left(\mathbf{h}_i^{v_1}, \mathbf{h}_t^{v_2}\right)/\tau\right)} \tag{19}$$

where τ is the temperature parameter that is fixed to 0.5. Compared with algorithms that only perform contrast loss on the identity matrix or adjacency matrix, our algorithm can capture long-distance positive and negative samples with their own information and semantic information.

F. Triplet Self-Supervised Clustering

The aforementioned graph contrastive learning module has the ability to recover discriminative and compact node embeddings from two graph augmentation inputs. To explicitly preserve and guide the consensus data cluster distribution between the two node embeddings, we propose a triplet self-supervised clustering module. In this module, a fusion embedding $\mathbf{H}^F$ is first constructed by combining both node embeddings

$$\mathbf{H}^F = \left[\mathbf{H}^{v_1}, \mathbf{H}^{v_2}\right]. \tag{20}$$

Specifically, based on the fusion embedding $\mathbf{H}^F$, we compute the soft distribution $\mathbf{Q}$ of pseudo-labels using the Student's t-distribution, i.e.,

$$q_{ij} = \frac{\left(1 + \left\|\mathbf{h}^F - \boldsymbol{\nu}_j\right\|^2/\eta\right)^{\left(-\frac{\eta+1}{2}\right)}}{\sum_{j=1}^{K}\left(1 + \left\|\mathbf{h}^F - \boldsymbol{\nu}_j\right\|^2/\eta\right)^{\left(-\frac{\eta+1}{2}\right)}} \tag{21}$$

where $\boldsymbol{\nu}_j$ is the jth cluster centroid, which is defined as the mean node embedding from the jth K-means label, and η represents the degrees of freedom. Similarly, based on the node embeddings $\mathbf{H}^{v_1}$ and $\mathbf{H}^{v_2}$, we can obtain the corresponding soft distributions $\mathbf{Q}^{v_1}$ and $\mathbf{Q}^{v_2}$.

To make the samples closer, we calculate the target distribution. The target distribution is computed as

$$p_{ij} = \frac{q_{ij}^2 / \sum_i q_{ij}}{\sum_{j'} q_{ij'}^2 / \sum_i q_{ij'}}. \tag{22}$$

The target distribution $\mathbf{P}$ can achieve a denser distribution by squaring $\mathbf{Q}$. After that, the triplet self-supervised clustering loss is formulated as

$$\begin{aligned}\mathcal{L}_{\mathrm{tsc}} &= \mathrm{KL}\left(\mathbf{P}\|\mathbf{Q}\right) + \mathrm{KL}\left(\mathbf{P}\|\mathbf{Q}^{v_1}\right) + \mathrm{KL}\left(\mathbf{P}\|\mathbf{Q}^{v_2}\right) \\ &= \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} + p_{ij} \log \frac{p_{ij}}{q_{ij}^{v_2}} + p_{ij} \log \frac{p_{ij}}{q_{ij}^{v_2}}\end{aligned} \tag{23}$$

where $\mathrm{KL}(\cdot\|\cdot)$ represents the Kullback–Leibler divergence between two distributions. Apart from pursuing the consensus data cluster distribution, the triplet self-supervised clustering loss can avoid over-reliance on information from a certain view and enhance the consistency of clustering structures of each view.

Algorithm 1 Learning Procedure for GCLHR

Input: Hyperspectral image data $\mathbf{X}$;
Number of neighbors k; Number of superpixels M;
Learning rate ξ; Number of training epochs I;
Momentum factor γ; Balance parameter β;
Network parameters $\{\boldsymbol{\Theta}_{\mathrm{VAE}}, \boldsymbol{\Omega}_{\mathrm{VAE}}\}, \{\boldsymbol{\Theta}_{\mathrm{GAE}}^{v_1}, \boldsymbol{\Theta}_{\mathrm{GAE}}^{v_2}\}$.
Output: Clustering results

```
Initialize the network parameters {Θ_VAE, Ω_VAE} and {Θ^{v1}_GAE, Θ^{v2}_GAE} of VAE and GAE;
Pre-training and Pre-processing:
Obtain the pixel cubes P;
Perform pre-training by minimizing Eq. (6);
Obtain superpixel segmentation result by Eq. (7)-Eq. (10);
Construct adjacency matrices A^{v1} and A^{v2};
Fine-tuning and Optimization:
for i = 1 to ≤ I do
    Obtain features H^{v1} and H^{v2} by Eq. (13) and Eq. (14);
    Calculate contrastive loss L_con by Eq. (15)-Eq. (19);
    Calculate self-supervised loss L_tsc by Eq. (20)-Eq. (23);
    Calculate loss L_FT = L_con + β * L_tsc;
    Minimize L_FT using stochastic gradient descent;
    Update Θ^{v1}_GAE = Θ^{v1}_GAE − ξ∇_{Θ_GAE} L_FT;
    Update Θ^{v2}_GCN ← γΘ^{v1}_GCN + (1 − γ)Θ^{v2}_GCN.
end for
return: Clustering result
```

G. Model Optimization

The GCLHR we proposed is a two-stage algorithm. In the pretraining stage, we only need the loss $\mathcal{L}_{\mathrm{pre}}$ to optimize the model. Then, we use the total loss

$$\mathcal{L}_{\mathrm{FT}} = \mathcal{L}_{\mathrm{chr}} + \beta * \mathcal{L}_{\mathrm{tsc}} \tag{24}$$

to optimize the model, where β is the balance parameter.

The three important parameters that need to be optimized for the VAE and GCN models are $\{\boldsymbol{\Theta}_{\mathrm{VAE}}, \boldsymbol{\Omega}_{\mathrm{VAE}}, \boldsymbol{\Theta}_{\mathrm{GCN}}^{v_1}, \boldsymbol{\Theta}_{\mathrm{GCN}}^{v_2}\}$. Given the learning rate ξ, the network parameters are updated by

$$\boldsymbol{\Theta}_{\mathrm{VAE}}^{*} = \boldsymbol{\Theta}_{\mathrm{VAE}} - \frac{\xi}{N}\sum_{i=1}^{N}\left(\frac{\partial \mathcal{L}_{\boldsymbol{\mu},\boldsymbol{\sigma}^2}}{\partial \boldsymbol{\Theta}_{\mathrm{VAE}}} + \frac{\partial \mathcal{L}_{\mathrm{rec}}}{\partial \boldsymbol{\Theta}_{\mathrm{VAE}}}\right) \tag{25}$$

$$\boldsymbol{\Omega}_{\mathrm{VAE}}^{v_1 *} = \boldsymbol{\Omega}_{\mathrm{VAE}}^{v_1} - \frac{\xi}{N}\sum_{i=1}^{N}\left(\frac{\partial \mathcal{L}_{\boldsymbol{\mu},\boldsymbol{\sigma}^2}}{\partial \boldsymbol{\Omega}_{\mathrm{VAE}}} + \frac{\partial \mathcal{L}_{\mathrm{rec}}}{\partial \boldsymbol{\Omega}_{\mathrm{VAE}}}\right) \tag{26}$$

$$\boldsymbol{\Theta}_{\mathrm{GCN}}^{v_1 *} = \boldsymbol{\Theta}_{\mathrm{GCN}}^{v_1} - \frac{\xi}{M}\sum_{i=1}^{M}\left(\frac{\partial \mathcal{L}_{\mathrm{con}}}{\partial \boldsymbol{\Theta}_{\mathrm{GCN}}^{v_1}} + \beta\frac{\partial \mathcal{L}_{\mathrm{tsc}}}{\partial \boldsymbol{\Theta}_{\mathrm{GCN}}^{v_1}}\right). \tag{27}$$

To ensure the stability of model optimization, further increase the disparity between the output features of the two branches, and enhance the optimization of contrastive learning, we adopt the exponential moving average (EMA) strategy to optimize the parameter $\boldsymbol{\Theta}_{\mathrm{GCN}}^{v_2}$

$$\boldsymbol{\Theta}_{\mathrm{GCN}}^{v_2} \leftarrow \gamma\boldsymbol{\Theta}_{\mathrm{GCN}}^{v_1} + (1-\gamma)\,\boldsymbol{\Theta}_{\mathrm{GCN}}^{v_2} \tag{28}$$

where γ is the momentum factor that has been fixed as 0.99. The detailed learning procedure of GCLHR is shown in Algorithm 1.

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IV. Experiments

A. Experimental Setup

1) Datasets: To rigorously evaluate the proposed methodology, four distinct hyperspectral datasets are utilized. Each dataset has unique spectral, spatial, and land-cover characteristics, thereby providing a diverse set of classification challenges. Below is the basic information of the four datasets.

a) Pavia University: The Pavia University dataset originates from an HSI acquired in 2003 by ROSIS-03, operated on a German airborne platform over the city of Pavia in northern Italy. This sensor initially recorded 115 continuous bands ranging from 0.43 to 0.86 μm, but 12 bands severely affected by noise were discarded, leaving 103 usable bands. The spatial coverage measures 610×340 pixels, with a ground sampling distance of roughly 1.3 m.

b) Salinas: The Salinas dataset, captured by an AVIRIS sensor in California's Salinas Valley, contains 204 spectral bands spanning 512 rows and 217 columns, with a spatial resolution of 3.7 m/pixel. It covers various agricultural land uses and includes 16 ground-truth classes.

c) Trento: The Trento dataset, acquired over northern Italy, offers high 1-m spatial resolution and 63 spectral bands. Covering 600×166 pixels, it includes six main classes, from apple orchards to built-up areas, representing a mix of rural and semi-urban land covers.

d) Xuzhou: The Xuzhou dataset, captured in November 2014 using a HYSPEX hyperspectral camera over peri-urban Xuzhou, China, consists of 436 spectral bands (after removing noisy channels) and covers 500×260 pixels with a spatial resolution of 0.73 m/pixel. It includes nine classes, such as crops, vegetation, man-made surfaces, and coal mining areas.

2) Compared Methods: To show the effectiveness of our proposed GCLHR, we compare it with 17th clustering methods, including the distance-based algorithms (K-means [15] and FCM [16]), sparse subspace clustering (SSC) [21], autoencoder-based deep clustering (AEC) [58], deep encoder-based clustering (DEC) [59], deep clustering with local structure preservation (IDEC) [60], selective sampling-based scalable SSC (S^5C) [22], hierarchical subspace clustering algorithm (HESSC) [23], multiscale spectral–spatial association network (MS^2A-Net) [61], dual graph autoencoder-based clustering method (DGAE) [29], neighborhood contrastive subspace clustering network (NCSC) [35], adaptive graph convolutional subspace clustering algorithm (AGCSC) [46], deep spectral clustering with regularized linear embedding (DSCRLE) [62], pixel-superpixel contrastive learning method (PSCPC) [33], enhanced multiview contrastive learning method (EMVCC) [34], multiview graph contrastive learning algorithm (CMSCGC) [36], and self-supervised double-structure transformer model (SDST) [26].

3) Implementation Details: The main parameters of GCLHR are carefully fine-tuned to obtain optimal performance and are used in subsequent experiments. The model is trained for 200 epochs, with each experiment repeated $10\times$, and the final results averaged to minimize the effects of randomness. We set the learning rate to 1×10^{-5} and employ the Adam optimizer for training. Both branches of the graph neural network encoder use a three-layer GCN, with hidden and output dimensions of 1024, 1024, and 512. Each of the first two GCN layers is directly followed by a ReLU activation and a dropout layer. To ensure fair comparisons between the proposed GCLHR and other baselines under the same experimental conditions, we performed ten runs on a system equipped with a 24-GB RTX 3090 GPU and 64 GB of RAM and then computed the average results. For each baseline method, we used the optimal parameters reported in the related literature and the corresponding open-source code. Due to the limitations of AGCSC and CMSCGC on large-scale datasets, we introduced a data preprocessing step that employs superpixels as their input. To clearly demonstrate the performance, we employ three evaluation metrics: overall accuracy (ACC), Kappa, and normalized mutual information (NMI) [63], [64], [65], [66], [67], [68], [69].

B. Experimental Results

Table I presents the clustering results of each algorithm on the Salinas, Pavia University, Trento, and Xuzhou datasets. Compared with other comparison algorithms, the GCLHR algorithm proposed in this article achieved the highest accuracy on these four datasets, reaching 85.32%, 64.11%, 85.07%, and 78.53%. This indicates that the proposed method has achieved the best clustering effect. Additionally, we can observe the following phenomena.

Traditional clustering methods such as K-means, FCM, and SSC performed poorly. To some extent, this reflects that traditional clustering algorithms are not suitable for complex and nonlinear HSI data. Among them, K-means performed the worst because this algorithm is suitable for sparse low-dimensional data and cannot handle high-dimensional nonsparse data well, making it difficult to identify nonspherical clusters. In contrast, the FCM algorithm performs better than K-means as it is a soft clustering algorithm that can improve clustering accuracy to a certain degree. The SSC algorithm performs slightly better because it is a clustering algorithm based on a graph structure, which can better adapt to the characteristics of data distribution, especially for complex nonspherical cluster structures. However, due to its single structure, it is difficult for it to extract the deep features of the data. Compared with clustering methods based on deep networks, it is still challenging to obtain good experimental results.

Deep network-based algorithms such as AEC, DEC, IDEC, and MS^2A-Net generally perform better. They leverage the powerful nonlinear transformation ability and dimensionality reduction technology of deep neural networks to cluster high-dimensional nonlinear data, and their experimental results are superior to those of traditional clustering algorithms. More importantly, deep network-based clustering algorithms can learn the deep feature representation in HSI, providing a good foundation for subsequent clustering. However, our proposed GCLHR still achieves the best performance.

Compared with the CMSCGC method, which is also based on graph contrast learning, our method improves the accuracy by 13.16%, 6.18%, 3.0%, and 6.47% on the four datasets. The significant improvement in accuracy is because we use

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TABLE I

CLUSTERING RESULTS ON FOUR HSI DATASETS. THE BEST AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED WITH BOLD AND UNDERLINE, RESPECTIVELY

Methods	Salinas			Pavia University			Trento			XuZhou		
	ACC	Kappa	NMI	ACC	Kappa	NMI	ACC	Kappa	NMI	ACC	Kappa	NMI
K-means (TPAMI'02) [15]	0.6111	0.5758	0.7124	0.5174	0.4199	0.5214	0.6281	0.5296	0.4618	0.4042	0.2958	0.4240
FCM (SSBM'13) [16]	0.5598	0.5226	0.6877	0.4273	0.3379	0.4995	0.5068	0.4127	0.4618	0.3991	0.3132	0.4224
SSC (TAPMI'13) [21]	0.6546	0.6234	0.7647	0.5204	0.4157	0.4213	0.7659	0.6965	0.6692	0.6974	0.6323	0.5758
AEC (ICML'12) [58]	0.6354	0.5932	0.7677	0.4575	0.3525	0.5176	0.7756	0.7085	0.7720	0.6666	0.5909	0.6497
DEC (ICML'16) [59]	0.5722	0.5146	0.7278	0.4514	0.3522	0.5401	0.6488	0.5366	0.6600	0.4965	0.3734	0.4240
IDEC (IDEC'17) [60]	0.6221	0.5791	0.7550	0.4479	0.3295	0.4848	0.7299	0.6603	0.7857	0.6775	0.6035	0.6584
S^{5}C (NeurIPS'19) [22]	0.5351	0.4934	0.6208	0.4836	0.3773	0.4323	0.4742	0.3219	0.3868	0.6314	0.5504	0.5226
HESSC (RS'20) [23]	0.5293	0.4787	0.6003	0.4771	0.3626	0.4943	0.4994	0.3031	0.3283	0.4235	0.2669	0.3242
MS^{2}A-Net (JSTARS'22) [61]	0.6463	0.6031	0.7447	0.6040	0.4815	0.5336	0.8360	0.7814	0.7624	0.6261	0.5377	0.5821
DGAE (TCSVT'22) [29]	0.5950	0.5560	0.7089	0.5265	0.4289	0.4755	0.7985	0.7300	0.6588	0.6761	0.5993	0.6105
NCSC (TGRS'22) [35]	0.7497	0.7207	0.7969	0.5107	0.4211	0.4531	0.6685	0.5823	0.6307	0.7403	0.6833	0.7239
AGCSC (CVPR'23) [46]	0.6384	0.5978	0.7665	0.5763	0.4699	0.4760	0.6552	0.5663	0.5913	0.7607	0.7039	0.6952
DSCRLE (TGRS'23) [62]	0.5713	0.5289	0.6086	0.4850	0.3654	0.4492	0.7711	0.6904	0.6569	0.7106	0.6496	0.6228
PSCPC (ICASSP'24) [33]	0.6519	0.6021	0.6108	0.5936	0.4516	0.5044	0.7826	**0.8244**	0.7435	0.7414	0.6786	0.6848
EMVCC (ACM MM'24) [34]	0.7709	0.7312	0.8379	0.6337	0.5471	0.5445	0.8003	0.7446	0.7139	0.7506	0.6975	0.7323
CMSCGC (TGRS'24) [36]	0.7216	0.6629	0.6916	0.5793	0.4346	0.4919	0.8477	0.8190	0.7988	0.7206	0.6662	0.6601
SDST (TGRS'24) [26]	0.7528	0.7507	0.8352	0.5070	0.3917	0.4266	0.8296	0.7655	0.7192	0.6700	0.5772	0.6282
Ours	**0.8532**	**0.8363**	**0.8825**	**0.6411**	**0.5656**	**0.6154**	**0.8507**	0.8019	**0.8033**	**0.7853**	**0.7369**	**0.7734**

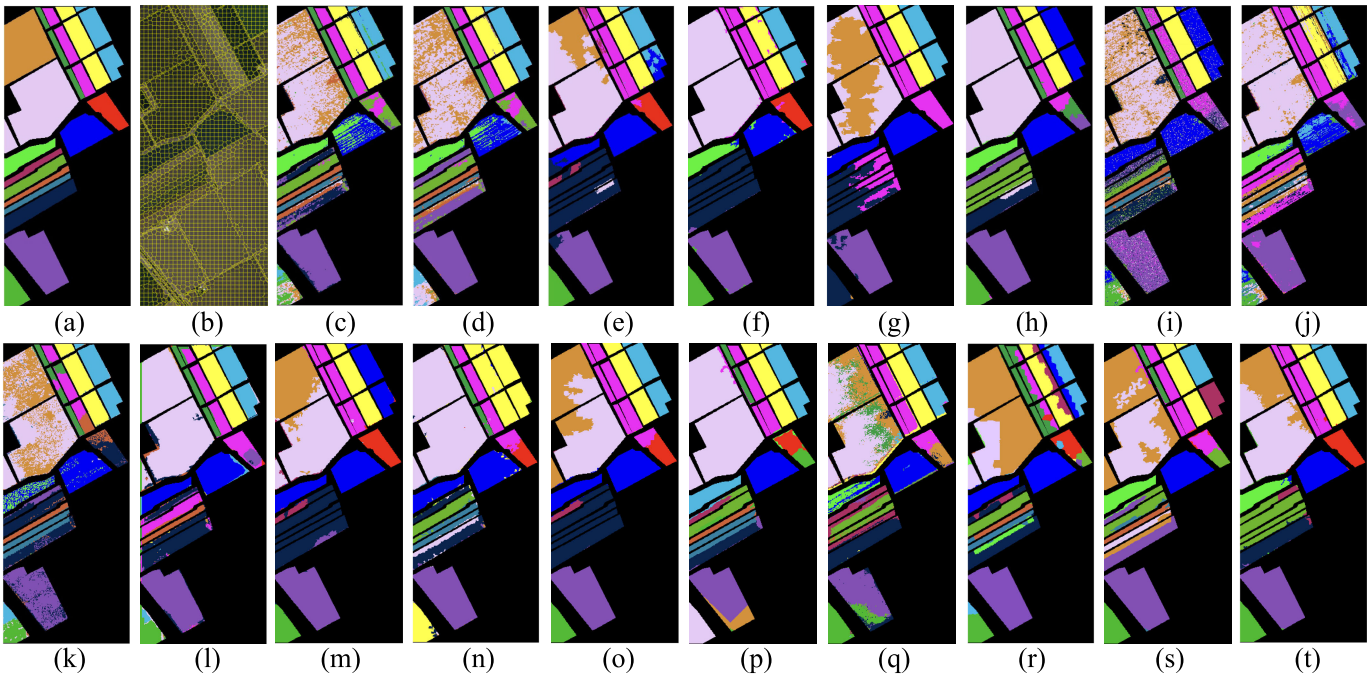


Fig. 3. Clustering maps on the Salinas dataset. (a) Ground truth, (b) superpixel segmentation visualization, (c) K-means, (d) FCM, (e) SSC, (f) AEC, (g) DEC, (h) IDEC, (i) S^{5}C, (j) HESSC, (k) MS^{2}A-Net, (l) DGAE, (m) NCSC, (n) AGCSC, (o) DSCRLE, (p) PSCPC, (q) EMVCC, (r) CMSCGC, (s) SDST, and (t) our proposed GCLHR.

high-order random walks to enhance contrastive learning. Compared with CMSCGC, which only uses one sample from the additional view as a positive sample, our method can effectively reduce the problem of incorrect sampling. Moreover, our triplet self-supervision algorithm and the semantic enhancement-based data augmentation method can provide good support for clustering tasks.

C. Qualitative Comparison

In order to provide an intuitive comparison of the effectiveness of the proposed GCLHR, we visualized the clustering results produced by various algorithms on three different datasets, as shown in Figs. 3–5. From these visualizations, it becomes apparent that conventional clustering methods often generate numerous misclassified pixels, leading to substantial

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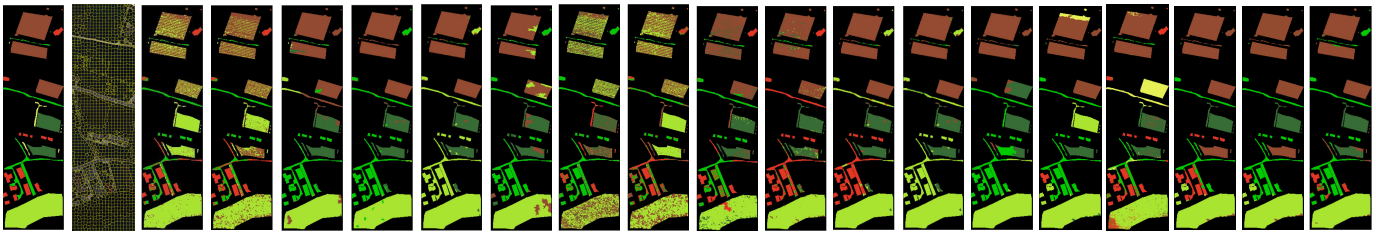


Fig. 4. Clustering maps on the Trento dataset. (a) Ground truth, (b) superpixel segmentation visualization, (c) K-means, (d) FCM, (e) SSC, (f) AEC, (g) DEC, (h) IDEC, (i) S^5C, (j) HESSC, (k) MS^2A-Net, (l) DGAE, (m) NCSC, (n) AGCSC, (o) DSCRLE, (p) PSCPC, (q) EMVCC, (r) CMSCGC, (s) SDST, and (t) our proposed GCLHR.

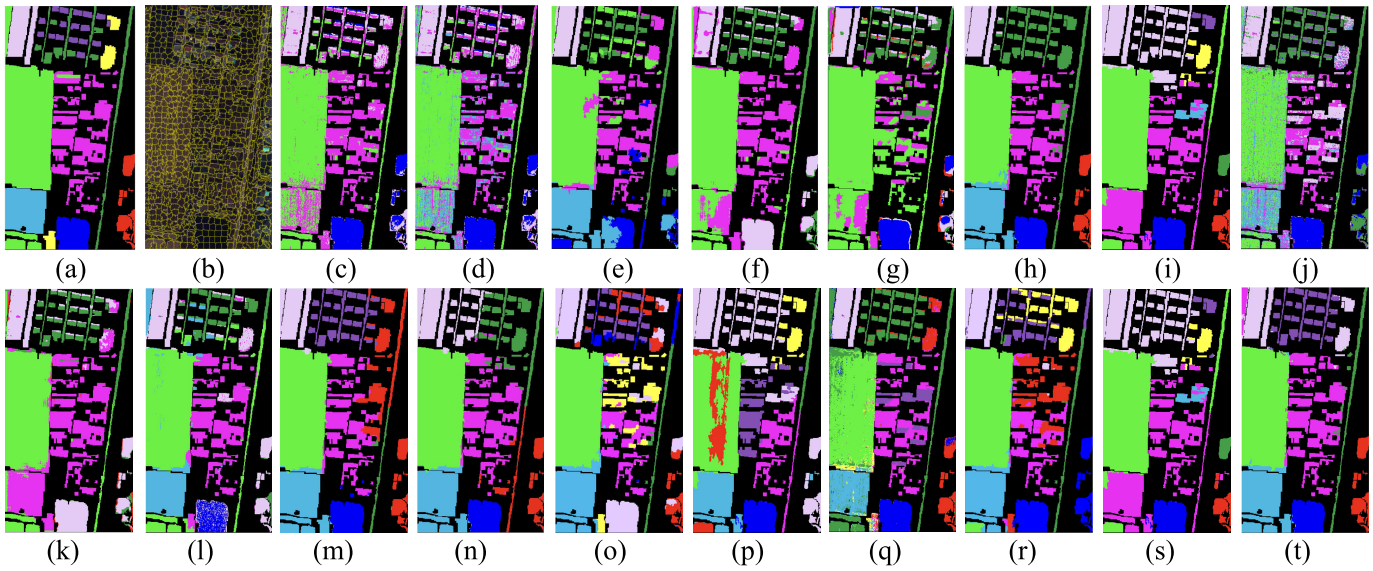


Fig. 5. Clustering maps on the Xuzhou dataset. (a) Ground truth, (b) superpixel segmentation visualization, (c) K-means, (d) FCM, (e) SSC, (f) AEC, (g) DEC, (h) IDEC, (i) S^5C, (j) HESSC, (k) MS^2A-Net, (l) DGAE, (m) NCSC, (n) AGCSC, (o) DSCRLE, (p) PSCPC, (q) EMVCC, (r) CMSCGC, (s) SDST, and (t) our proposed GCLHR.

noise in the resulting images. Approaches such as K-means, FCM, S5C, and HESSC commonly exhibit noticeable salt-and-pepper noise, largely due to insufficient exploitation of spatial correlations in the data. Consequently, their clustering maps tend to suffer from reduced continuity and overall quality.

In contrast, large-scale HSI clustering methods generally yield smoother clustering results, primarily because they more comprehensively leverage both spectral and spatial information. For instance, algorithms that integrate graph convolution (e.g., DGAE and AGCSC) capture spatial relationships more effectively, thereby producing smoother clustering maps and mitigating the impact of noise on visualization. Ultimately, experimental results on all four datasets demonstrate that GCLHR can clearly and accurately distinguish most pixel categories with minimal noise, thus delivering the sharpest visual quality. Notably, GCLHR also excels at preserving edge details, which is particularly crucial in scenarios requiring precise boundary delineation. For example, on the Salinas dataset, the proposed method provides smoother clusters for the "Grapes" and "Vinyard" categories. Overall, these findings indicate that GCLHR effectively mitigates both false positives and false negatives, thereby achieving significantly improved clustering accuracy and more reliable visual outcomes.

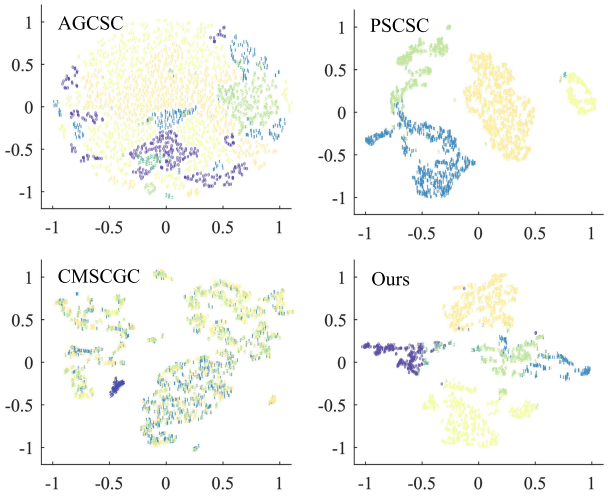


Fig. 6. t-SNE visualization results on the Trento dataset.

Fig. 6 presents the t-SNE visualization results on the Trento dataset, where the clusters produced by our method exhibit remarkably high intracluster compactness. Data points are highly concentrated in the low-dimensional space, and the

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TABLE II
RUNNING TIME (SECOND) OF DEEP LEARNING-BASED METHODS ON FOUR DATASETS

Dataset	AE	DEC	IDEC	HESSC	MS^{2}A-Net	DGAE	NCSC	DSCRLE	PSCPC	EMVCC	CMSCGC	SDST	Ours
Salinas	949	794	586	730	488	46	500	125	92	405	622	986	204
PaviaU	824	647	373	1014	152	43	621	235	190	232	735	925	294
Trento	524	443	190	555	106	39	435	124	86	464	831	552	74
XuZhou	1113	950	754	960	473	117	420	538	101	546	573	1842	68

within-cluster density is substantially higher than that of AGCSC, PSCSC, and CMSCGC. In contrast, AGCSC and CMSCGC show noticeable dispersion, with portions of data points scattered away from the main cluster body, resulting in loose cluster structures and blurred boundaries. PSCPC achieves a moderate improvement but still displays a number of points deviating from the core cluster region. This pronounced compactness indicates that our method captures the intrinsic characteristics of the data more precisely, effectively suppresses redundant information, and promotes more consistent feature representations for samples within the same class.

D. Running Time

To evaluate the complexity of different algorithms and highlight the runtime efficiency of our proposed GCLHR, we compared its execution time against several other deep learning-based HSI clustering methods across four datasets, as shown in Table II. The results indicate a distinct advantage for our model, particularly on the Xuzhou dataset, where it achieves the shortest running time. While algorithms such as AE and SDST require thousands of seconds to complete clustering on Xuzhou, our approach reduces this time to just 68 s. Similarly, on the Trento dataset, AE and HESSC take over 500 s, whereas our method only needs 74 s.

One reason for the slower performance of algorithms like AE, DEC, and IDEC is that they operate at the pixel level, leading to higher computational overhead. Subspace clustering methods such as NCSC and CMSCGC also incur considerable time costs due to their complex optimization processes, which, while potentially improving accuracy, demand significant computational resources. In contrast, our optimization strategy lowers runtime while boosting accuracy, underscoring its efficiency and scalability across diverse datasets. Although GCLHR's total running time is marginally longer than that of DGAE—the fastest method in our comparison—the notable gains in clustering accuracy justify this additional computational expense. Overall, GCLHR strikes an effective balance between accuracy and runtime, demonstrating exceptional efficiency and rendering it well-suited for real-world HSI analysis tasks.

E. Ablation Study

We carry out ablation experiments from six aspects: only the pretraining module, without the designed heterogeneous region generation module but simply averaging the pixels to obtain superpixel features (w/o HRG), with the proposed semantic information-based graph augmentation module (w/o SIGA), without the graph contrastive learning loss $\mathcal{L}_{\text{chr}}$ based on high-order relations, and without the triplet self-supervised learning loss $\mathcal{L}_{\text{tsc}}$. The importance of the pretraining module based on the 3-D encoder is investigated by keeping only the pretrained part. Meanwhile, the impact on the clustering performance is tested by removing the proposed contrastive learning loss and self-supervised loss. The results are summarized in Table III.

TABLE III
ABLATION RESULTS OF DIFFERENT VARIANTS

Datasets	Variants	ACC	Kappa	NMI
Salinas	Pre-training	0.7093	0.6714	0.6915
	w/o HRG	0.8414	0.8231	0.8787
	w/o SIGA	0.8408	0.8223	0.8816
	w/o $\mathcal{L}_{\text{chr}}$	0.7783	0.7549	0.8176
	w/o $\mathcal{L}_{\text{tsc}}$	0.8293	0.8109	0.8745
	Ours	**0.8532**	**0.8363**	**0.8825**
Pavia University	Pre-training	0.5573	0.4738	0.6021
	w/o HRG	0.6373	0.5604	0.6189
	w/o SIGA	0.6254	0.5437	0.6028
	w/o $\mathcal{L}_{\text{chr}}$	0.6159	0.5351	**0.6296**
	w/o $\mathcal{L}_{\text{tsc}}$	0.6254	0.5463	0.6141
	Ours	**0.6411**	**0.5656**	0.6154
Trento	Pre-training	0.7007	0.6308	0.7226
	w/o HRG	0.8385	0.7872	0.8022
	w/o SIGA	0.8342	0.7816	0.7957
	w/o $\mathcal{L}_{\text{chr}}$	0.8095	0.7211	0.7761
	w/o $\mathcal{L}_{\text{tsc}}$	0.8261	0.7620	0.7864
	Ours	**0.8507**	**0.8019**	**0.8033**
XuZhou	Pre-training	0.6759	0.6052	0.6691
	w/o HRG	0.7661	0.7146	0.7694
	w/o SIGA	0.7500	0.7106	0.7415
	w/o $\mathcal{L}_{\text{chr}}$	0.7279	0.6674	0.7345
	w/o $\mathcal{L}_{\text{tsc}}$	0.7246	0.6634	0.7269
	Ours	**0.7853**	**0.7369**	**0.7734**

Based on the experimental results, it can be observed that merely using autoencoders to extract deep features fails to achieve outstanding clustering performance. However, when combined with the K-means clustering results in Table I, it is evident that the deep features extracted by the pretraining module still contribute to improved clustering results for the model. Specifically, the improvements are 9.82%, 3.99%, 7.26%, and 27.17% on the four datasets, thus laying a reliable foundation for the subsequent view augmentation and superpixel feature construction modules based on semantic information.

As shown in the results, across all four datasets, the full model consistently outperforms the variant without SIGA (w/o SIGA). The ACC improvements are 1.40%, 1.57%, 1.65%, and 3.53%, demonstrating the stable and generalizable positive contribution of the SIGA module. The reason is that,

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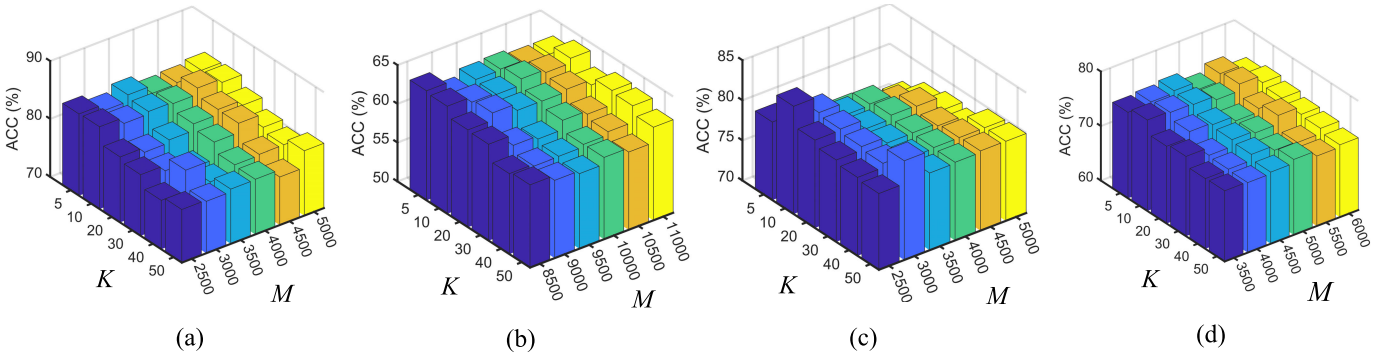


Fig. 7. 3-D visualization of GCLHR clustering results on (a) Salinas, (b) Pavia University, (c) Trento, and (d) Xuzhou datasets with different k and M values.

compared with traditional KNN-based graph construction, SIGA leverages K-means clustering to capture global semantic relationships among samples and refines the adjacency matrix through the intersection of multiple clustering results. This process effectively mitigates the noise introduced by random edge perturbations, resulting in a graph structure that more faithfully reflects the true class distribution of the data.

When superpixel features are constructed using only simple pixel feature averaging (w/o HGR), the model performance decreases by 1.18%, 0.38%, 1.22%, and 1.92% on the four datasets. These results indicate that the proposed distance-based heterogeneous region generation module contributes effectively to performance improvement. Specifically, we intentionally preserve a positive correlation between distance and weight to enhance the response to pixels located along superpixel boundaries, thereby enabling the aggregated superpixel features to more accurately capture boundary variations. By emphasizing these edge pixels, the model becomes more sensitive to boundary information and better characterizes the spatial heterogeneity within each superpixel.

Furthermore, after removing the graph contrastive learning loss $\mathcal{L}_{\text{chr}}$ based on high-order relations and the triplet self-supervised contrastive learning, the model performance drops by 7.49%/2.39%, 2.52%/1.57%, 4.12%/2.46%, and 5.74%/6.07%. This indicates that the losses we designed can significantly enhance the model performance. This is also because the graph contrastive learning module can measure the similarity between samples, thereby capturing distant positive samples for each node and reducing the interference of incorrect negative samples. Meanwhile, the triplet self-supervised loss can facilitate the consistency of learning between views and prompt the model to jointly learn features that are favorable for clustering.

F. Hyperparameter Analysis

1) Impact of Hyperparameters k and M: In the GCLHR method, the hyperparameter k controls the number of neighbors during graph construction, while M determines the number of superpixels. To investigate the impact of different values of k and M on the model, we designed an experiment where other parameters were kept constant. The range for k was set to 5, 10, 20, 30, 40, and 50. For the Salinas dataset, M was varied from 2500 to 5000; for the Pavia University dataset, M ranged from 8500 to 11 000; for the Trento dataset, M was set between 2500 and 5000; and for the Xuzhou dataset, M was adjusted from 3500 to 6000, with a step size of 500. The results shown in Fig. 7 indicate that when k is too small, the model can only leverage limited neighborhood information, making it difficult to capture richer spatial features. Conversely, when k is too large, excessive noise may be introduced, leading to oversmoothing and a decrease in model performance. Similarly, if the number of superpixels M is too small, overaggregation occurs, causing many local details to be lost; if M is too large, the number of nodes in the graph increases significantly, raising computational costs and potentially amplifying noise. Overall, as k and M were gradually increased, the model typically achieved the best performance in intermediate or slightly smaller value ranges, indicating the need to balance neighborhood size and superpixel granularity. This balance helps retain essential details while avoiding excessive smoothing or noise accumulation, ultimately leading to better overall accuracy and lower computational complexity. Therefore, for these four datasets, we selected the following settings for (k, M): (10, 4500), (5, 8500), (10, 2500), and (10, 3500).

2) Impact of Hyperparameters β: The self-supervised loss is a crucial component of the total loss function in GCLHR, with its weight controlled by the parameter β. We set the range of β to the following values: 0.0001, 0.001, 0.01, 0.1, 1, 10, 100, and 1000, and used four evaluation metrics on the four datasets as performance criteria. The final results are shown in Fig. 8. It can be observed that our method achieves relatively stable performance within a certain range of β, indicating that GCLHR is not highly sensitive to this parameter on some datasets. On the Trento dataset, however, the value of β has the greatest impact on the model's performance, which may be due to the dataset's low homogeneity and the presence of more challenging samples.

3) Impact of Hyperparameters t: The hyperparameter t determines the number of random walks, with larger values enabling the model to capture more extensive long-range dependencies. Fig. 9 reports the model performance across different values of t. As shown, performance generally improves with increasing t since distant samples are progressively integrated into the matrix, allowing the model to construct a greater number of informative positive and negative pairs.

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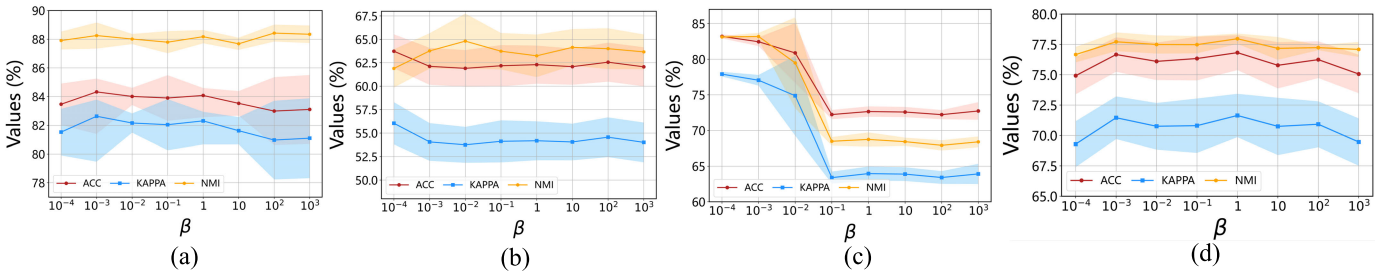


Fig. 8. Visualization of GCLHR clustering results on (a) Salinas, (b) Pavia University, (c) Trento, and (d) Xuzhou datasets with different parameters β.

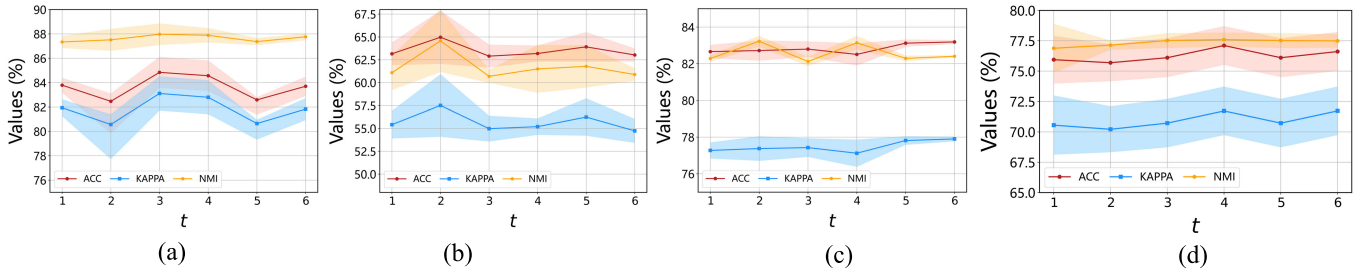


Fig. 9. Visualization of GCLHR clustering results on (a) Salinas, (b) Pavia University, (c) Trento, and (d) Xuzhou datasets with different parameters t.

Nevertheless, when t becomes excessively large, performance deteriorates, presumably because extended spatial relationships introduce noisy or irrelevant samples that hinder effective training. Consequently, it is essential to select t within a moderate range to achieve a balance between capturing long-distance relationships and suppressing noise.

V. Conclusion

In this article, a novel graph contrastive learning method with high-order relations is put forward to address the issue of incorrect capturing of positive and negative samples in the HSI clustering task. Specifically, we initially utilize a 3-D autoencoder to extract deep features and devise a distance-based superpixel construction method to obtain superpixel features, which will serve as the input for the GCN. Subsequently, we employ multiple random walks to construct an affinity graph that represents the similarity between each sample, with this being the objective of contrastive learning. This approach enables us to capture as many distant positive samples as possible, and at the same time, a lower similarity can filter out certain negative samples. Moreover, we design a triplet self-supervised loss to enhance the consistency between the two branches. Experiments carried out on multiple HSI datasets demonstrate that our proposed GCLHR method attains excellent and reliable clustering performance.

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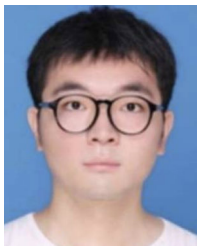
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